

AI/NLP Engine for SIF Precursor Detection in Oil & Gas UA/UC & Near-Miss Reports

Comprehensive Research Synthesis

Role: Lead AI/NLP Researcher & Principal Data Scientist — Industrial HSE, Oil & Gas
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1. Strong Fundamentals & Domain Context

1.1 Defining SIF Precursors vs. Standard Safety Observations

The conceptual distinction between a routine safety observation and a **Serious Injury or Fatality (SIF) Precursor** is the most critical definitional boundary in this entire project. Misclassification at this definitional level cascades into faulty annotation, wrong model objectives, and ultimately, a system that fails where it matters most.

Standard Safety Observation (UA/UC)

A Unsafe Act (UA) or Unsafe Condition (UC) report that documents a deviation from safe practice or a substandard physical state that, under typical circumstances, would *not* directly lead to permanent disability or death. These are high-frequency, low-severity-potential observations.

Example: "Housekeeping issue near pump shed. Oil spill not cleaned. Slipping hazard identified. Corrected on the spot."

SIF Precursor (Definition)

A SIF Precursor is a specific observable hazardous condition, human behavior, system state, or combination thereof that — **absent immediate corrective intervention** — possesses a direct causal pathway to a Serious Injury or Fatality event. The differentiating criterion is **energy magnitude** and **exposure proximity**.

The DEKRA/EHS framework establishes two primary SIF Precursor categories:

Category	Description	O&G Examples
Exposure Precursor	Worker is in the line-of-fire of high-energy hazard	Working under suspended load, inside H2S exclusion zone without BA, standing in pinch-point zone of rotating equipment
Release Precursor	Critical control has failed or is absent, enabling energy release	LOTO not applied on live circuit, PTW issued for wrong equipment, atmospheric test not conducted before confined space entry

Operationalized SIF Precursor Criteria (for Annotators):

A report qualifies as a SIF Precursor if it satisfies **at least ONE** of the following:

1. **High-energy hazard exposure:** Uncontrolled energies $\geq$ threshold (e.g., electrical >50V unguarded, pressure vessel >15 PSI, height >1.8m without fall arrest)
2. **Critical procedure failure:** PTW violation, LOTO bypass, atmospheric testing skip, JSA not performed for non-routine high-risk task
3. **Toxic/flammable material loss of containment:** H2S above IDLH (50 ppm), hydrocarbon vapor cloud present with ignition source within proximity
4. **Structural/geotechnical critical signal:** Ground subsidence near well, casing failure, riser fatigue crack
5. **Simultaneous operations (SIMOPS) conflict:** Concurrent activities creating uncontrolled interaction risks (drilling + hot work on same deck)
6. **Critical equipment failure on safety-critical systems:** BOP function test failure, emergency shutdown valve stuck, gas detection system offline

1.2 Linguistic Characteristics of UA/UC & Near-Miss Reports in O&G

Understanding the *language* of these reports is essential before selecting or fine-tuning any NLP model. O&G safety reports have a highly distinctive and challenging linguistic profile:

Characteristic 1: Domain-Specific Jargon & Acronyms (Dense)

Acronym	Expansion	Significance to SIF Detection
LOTO	Lockout/Tagout	Isolation of hazardous energy — absence = SIF precursor
PTW	Permit to Work	Critical administrative control — violation = SIF precursor
JSA/JHA	Job Safety Analysis / Job Hazard Analysis	Risk assessment document — missing = elevated SIF potential
SIMOPS	Simultaneous Operations	High-complexity multi-team scenario = elevated SIF potential
BOP	Blowout Preventer	Critical well control equipment
H2S	Hydrogen Sulfide	Toxic gas — immediately dangerous to life
IDLH	Immediately Dangerous to Life and Health	Threshold for SIF potential

Acronym	Expansion	Significance to SIF Detection
MOC	Management of Change	Bypassed MOC = uncontrolled change = SIF precursor
TBT	Toolbox Talk	Pre-task safety briefing — absence for high-risk task is significant
SMP/SOP	Safe Method of Practice / Standard Operating Procedure	Procedural baseline
ESD	Emergency Shutdown	Safety-critical system
LEL	Lower Explosive Limit	Flammable gas threshold — exceedance = SIF precursor
BA / SCBA	Breathing Apparatus / Self-Contained BA	PPE for toxic atmospheres
WC	Well Control	Critical drilling safety function
ERP	Emergency Response Plan	

NLP Implication: A model that does not "understand" that "LOTO not applied" is semantically equivalent to "energy isolation procedure bypassed" will fail silently on real-world data.

Characteristic 2: Telegraphic Sentence Structure

Reports are rarely written in complete, grammatically coherent sentences. Reporters are typically field workers filing quickly under operational pressure:

✗ Standard prose:

"The worker was observed performing a hot work operation without first obtaining the required hot work permit, in close proximity to hydrocarbon-containing equipment."

✓ Typical field report:

"HW carried out nr HC equipment w/o PTW. No gas test done."

This creates significant challenges for tokenization, sentence boundary detection, and dependency parsing.

Characteristic 3: Implicit Consequences & Omitted Context

The *consequence pathway* — what *could have* happened — is rarely stated. The report documents the *observed fact*, not the risk inference. This is the semantic gap the model must bridge.

Observed: "Pressure relief valve found manual override in 'bypass' position."
 Implicit SIF path: Blocked PRV → pressure vessel overpressure → catastrophic rupture → multiple fatalities

Characteristic 4: Mixed Language & Regional Variation

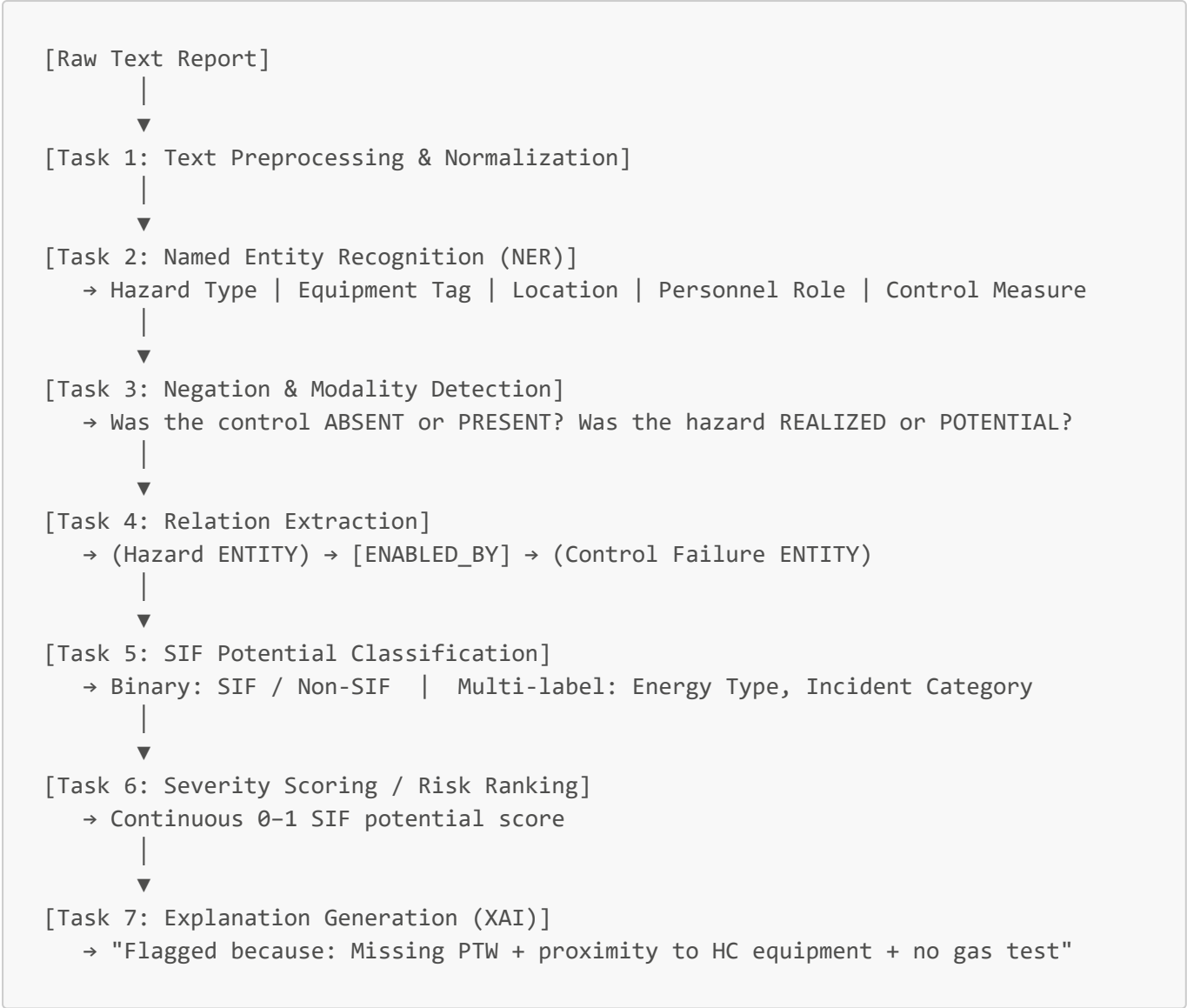
For OIL specifically (operating in Assam, Arunachal Pradesh, Rajasthan): reports may contain **code-switched Assamese-English or Hindi-English** phrases. Example: "*Kaam banda karna tha, but worker ne PTW nahi liya.*" This demands multilingual tokenization capabilities.

Characteristic 5: Noise Characteristics

- **Typographical inconsistency:** "LOTO", "L.O.T.O", "Lockout tagout", "lock-out/tag-out" all refer to the same concept
 - **Negation density:** "No PTW issued" vs "PTW not required as per supervisor" — both contain 'PTW' but carry opposite safety signals
 - **Temporal ambiguity:** "Recently observed", "during morning shift" — imprecise timing
 - **Passive blame-shifting language:** Reporters may deliberately obscure severity to protect colleagues or avoid bureaucratic escalation
-

1.3 Foundational NLP Tasks Required

The system is not a single NLP task — it is a **pipeline of composable NLP components**, each feeding the next:



2. Data Gathering: Where to Find the Data

2.1 Internal Sources (Primary Training Corpus)

These constitute the most valuable data because they are domain- and site-specific. Engage your EHS IT team and HSE management for access agreements.

Primary Internal Systems

System / Source	Data Type	Key Tables/Modules	Volume Estimate
SAP EHS / S4HANA EHS	Structured + unstructured	QMEL (notifications), QMFE (defect items), long-text fields	Varies — typically 5k–50k reports/year for a major operator
Enablon / Intelex / Cority	Structured + rich text	Incident forms, corrective actions, observation descriptions	Platform-dependent

System / Source	Data Type	Key Tables/Modules	Volume Estimate
Maximo (IBM)	Work orders with safety flags	Hazard fields, failure codes, narrative descriptions	High volume, but noisy
OIL's internal SMS portal	Custom EHS reports	Observation cards, investigation reports	Specific to OIL infrastructure
Physical Safety Observation Cards	Unstructured handwritten	Post-OCR text	Requires digitization pipeline
HSE Audit Reports (PDF)	Narrative text	Finding descriptions, recommendations	Lower volume, high information density
Toolbox Talk Records	Free-form text	Pre-task hazard discussions	Often inconsistently captured
Emergency Call Logs / Incident MOM	Unstructured narrative	Immediate incident descriptions	Very high SIF-signal density
PTW Rejection Logs	Structured + comments	Reason for rejection field	High SIF precursor concentration

Data Acquisition Priority Ranking for OIL:

PTW Rejection Logs > Near-Miss Investigation Reports > Emergency Call Logs > SAP EHS Notifications > Observation Cards

Data Request Template for Internal IT/EHS

When submitting a data extraction request, specify:

- Date range: Minimum **5 years** of historical data (10 years preferred)
- Fields required: `Report_ID`, `Date`, `Location_Code`, `Reporter_Role`, `Description_Text`, `Action_Taken_Text`, `Consequence_Potential` (if captured), `Incident_Type_Code`
- Exclude fields: `Reporter_Name`, `Employee_ID`, `Supervisor_Name`, `Witness_Names`
- Format: Flat CSV or JSON (avoid Excel for large volumes)

2.2 External & Public Datasets (Pre-training, Benchmarking, Augmentation)

These datasets serve three functions: (a) **domain language pre-training**, (b) **weak labeling bootstrap**, and (c) **external benchmark validation**.

Tier 1 — Direct O&G Safety Text (Highest Relevance)

Dataset	Source	Access	Size	SIF Relevance	Use Case
OSHA Severe Injury	data.osha.gov	Public API	~130,000+ records (2015–present)	High — amputation, hospitalization,	Pre-training, benchmark

Dataset	Source	Access	Size	SIF Relevance	Use Case
Reports (SIR)				fatality narratives	
BSEE Offshore Incident Reports	BSEE.gov	Public download	~5,000+ offshore-specific reports	Very High — O&G specific	Domain pre-training
PHMSA Incident Reports	phmsa.dot.gov/data	Public	~30,000+ pipeline/hazmat incidents	High — HC release incidents	Augmentation
CSB Investigation Reports	csb.gov/investigations	Public PDFs	~150 major incident reports, rich narrative	Very High — SIF-confirmed events	Gold-standard SIF examples
UK HSE RIDDOR Database	HSE.gov.uk	Public	Hundreds of thousands	High — comprehensive incident data	Pre-training
EPA RMP Accident History	epa.gov/rmp	Public	~10,000 chemical plant accidents	Medium — process safety	Augmentation

Tier 2 — Adjacent Industrial Safety Text (Transfer Learning)

Dataset	Source	Use Case
MSHA Metal/Nonmetal Accident Data	arlo.msha.gov	NLP pre-training on industrial safety narrative
NTSB Pipeline Accident Reports	ntsb.gov	Additional HC-related narratives
eMARS (EU Major Accident Reporting System)	emars.jrc.ec.europa.eu	European process industry incidents
ARIA Database (BARPI, France)	aria.developpement-durable.gouv.fr	Accidental Risk Information Analysis
NRC Licensee Event Reports (LERs)	nrc.gov/reading-rm	Methodology model — nuclear safety reporting NLP research is mature

Tier 3 — Academic & Synthetic Sources

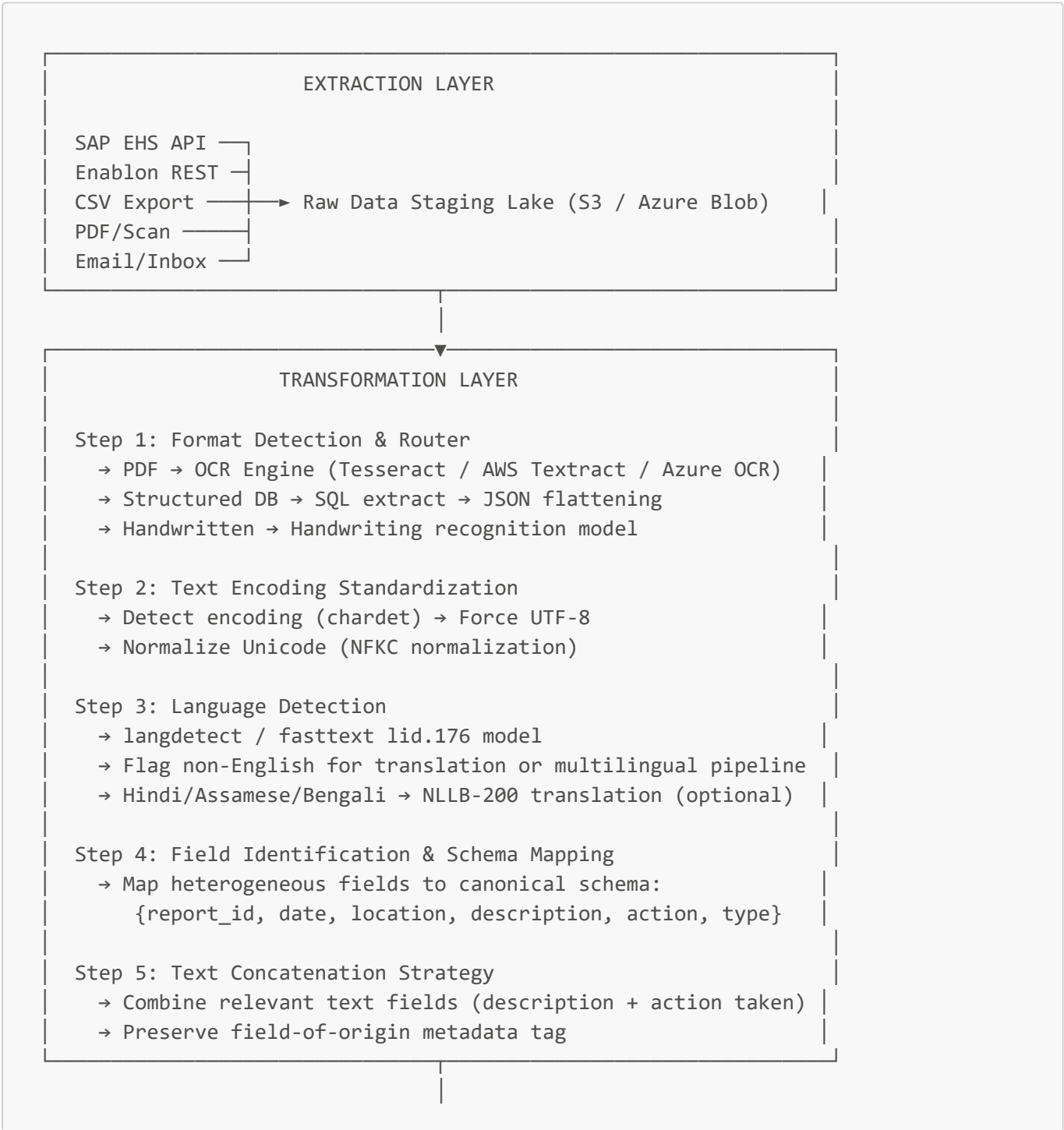
Source	Description
SSCI / Scopus / ScienceDirect	Search for: "NLP safety report classification", "incident text mining oil gas", "BERT safety incident" — for benchmark comparisons

Source	Description
AI-generated synthetic SIF reports	Use Claude/GPT-4 prompted with real incident archetypes to generate synthetic SIF precursor reports for augmentation (label with confidence = "synthetic")
LOTO/PTW vendor training materials	EHS training document corpora — useful for domain vocabulary pre-training

3. Data Curation: How to Gather & Prepare the Data

3.1 ETL Pipeline for Unstructured Safety Text

The extraction pipeline must handle heterogeneous source formats. Below is the recommended architecture:





OCR Quality Control for Handwritten/Scanned Reports:

Handwritten safety cards are common in field operations. Post-OCR, apply:

- Confidence threshold filtering: discard records with OCR confidence < 0.70
- Manual review queue for 0.50–0.70 confidence band
- Spell-check with O&G domain dictionary before further processing

3.2 Data Sanitization: PII Anonymization & Corporate Data Protection

This is **non-negotiable** and must be completed before any data leaves the secure corporate perimeter for model training.

PII Categories to Remove/Replace

PII Category	Detection Method	Replacement Strategy
Personal names (reporter, witness, supervisor)	spaCy NER (PERSON entity) + name gazetteer	Replace with role token: [WORKER_ROLE] , [SUPERVISOR]
Employee/Badge numbers	Regex: \b(EMP	ID
Phone/Contact numbers	Regex: international phone patterns	Replace with [CONTACT]
Email addresses	Regex: standard email pattern	Replace with [EMAIL]
Exact GPS coordinates	Regex + NER for coordinate patterns	Replace with zone code: [ZONE_NORTH_FIELD]
Contractor company names	Entity list + NER	Replace with [CONTRACTOR_ENTITY]
Witness statements with identifying detail	NER + context window review	Generalize or suppress
Injury body part details (quasi-identifier)	Medical NER (custom)	Generalize: right index finger → [EXTREMITY]

Recommended Toolchain for Anonymization

```
# Primary Stack
import spacy                # NER for PERSON, ORG, GPE entities
from presidio_analyzer import AnalyzerEngine    # Microsoft Presidio – PII
                                              detection
from presidio_anonymizer import AnonymizerEngine # Presidio anonymization engine
import re                   # Regex for structured PII patterns

# Domain-specific extension
# Custom NER model for O&G-specific quasi-identifiers
# (e.g., well names, production figures, specific incident IDs)
```

Corporate Sensitivity Layer (Beyond PII)

Beyond personal data, the following corporate-sensitive information must also be masked before external processing:

- **Well names and identifiers** → [WELL_ASSET]
- **Specific production volumes** mentioned in incident context → [PRODUCTION_DATA]
- **Financial loss estimates** → [FINANCIAL_DATA]
- **Specific equipment serial numbers / vendor names** → [EQUIPMENT_ID] / [VENDOR]
- **Regulatory citation references** that could identify pending investigations → [REG_REF]

Critical Recommendation: Establish a **Data Classification Committee** involving EHS, Legal, and IT Security to formally approve the anonymization schema *before* any data extraction begins. This prevents re-work and ensures PDPA/DPDP Act 2023 (India) compliance for OIL.

3.3 Handling Extreme Class Imbalance

In a typical O&G safety database, SIF precursors represent **2–8% of all UA/UC and near-miss reports**. This creates severe class imbalance that will cause a naïve model to achieve 95%+ accuracy by simply predicting "Non-SIF" for everything — a catastrophically dangerous outcome.

Imbalance Quantification: Establish First

Before applying any correction, measure:

- Class ratio: Expected 92–98% Non-SIF : 2–8% SIF
- Within SIF, sub-class frequencies (some SIF categories may be even rarer)
- Temporal distribution: Is imbalance stable across years?

Strategy 1 — Threshold Calibration (Always Apply)

Adjust the classification decision threshold post-training. Default threshold = 0.5 is inappropriate. Optimize threshold on validation set to maximize F2-score or achieve target Recall ≥ 0.90 .

```
from sklearn.metrics import f1_score, recall_score
# Sweep threshold
```

```
for threshold in np.arange(0.05, 0.95, 0.01):
    preds = (probabilities >= threshold).astype(int)
    f2 = fbeta_score(y_val, preds, beta=2)
    recall = recall_score(y_val, preds)
```

Strategy 2 — Cost-Sensitive Learning (Apply During Training)

Assign asymmetric class weights to the loss function. For a 95:5 ratio:

```
# PyTorch: Weighted BCE Loss
from torch.nn import BCEWithLogitsLoss
pos_weight = torch.tensor([19.0]) # 95/5 ratio
criterion = BCEWithLogitsLoss(pos_weight=pos_weight)

# Sklearn equivalent
from sklearn.utils.class_weight import compute_class_weight
weights = compute_class_weight('balanced', classes=[0,1], y=y_train)
```

Strategy 3 — Text Augmentation for SIF Class (Data-Level)

Generate additional SIF-positive training examples without collecting new data:

Technique	Method	Quality Level
Back Translation	English → Hindi → English via NLLB-200 or DeepL	Medium — preserves semantics, varies phrasing
Synonym Substitution	Replace non-critical words with domain synonyms via WordNet + domain thesaurus	Medium
LLM Paraphrase Generation	Prompt GPT-4/Claude to rephrase confirmed SIF reports with instruction to "preserve all safety-critical details"	High — best quality
Template-based Synthesis	Create SIF report templates from real incident archetypes; fill with variable equipment/location tokens	High — fully controlled
EDA (Easy Data Augmentation)	Random insertion, swap, deletion on non-critical tokens	Low — use sparingly

Caution: Never augment the test set. Augmentation is exclusively a training-set intervention.

Strategy 4 — Focal Loss for Transformer Training

```
# Focal Loss — down-weights easy negative examples
class FocalLoss(nn.Module):
```

```
def __init__(self, gamma=2.0, alpha=0.75):
    # alpha = weight for positive (SIF) class
    # gamma = focusing parameter (2.0 standard for imbalanced)
```

Strategy 5 — SMOTE-Analogues for Text (Advanced)

Traditional SMOTE operates in vector space. For text, apply after embedding:

- Generate TF-IDF or sentence-transformer embeddings of SIF reports
- Apply SMOTE in embedding space → generate synthetic feature vectors
- Use nearest-neighbor retrieval to reconstruct approximate text (or use directly as embedding-level features)

4. Data Annotation & Utilization: How to Use the Data

4.1 Labeling Taxonomy for SME Annotators

A well-defined taxonomy is the backbone of annotation quality. Below is a **hierarchical 4-level taxonomy** designed for O&G SIF precursor annotation:

Level 1 — Primary SIF Label (Binary, Mandatory)

Label	Code	Definition
Non-SIF Potential	0	Report describes a hazard/deviation that, under realistic worst-case conditions, would NOT directly cause permanent disability or fatality
SIF Potential	1	Report describes a situation that has a direct causal pathway to a fatal or permanent disabling injury if not immediately corrected

Level 2 — SIF Energy/Hazard Category (Multi-label, Mandatory if L1=1)

Code	Category	O&G Examples
SIF-E	Electrical High Energy	Live work without isolation, arc flash potential, substation intrusion
SIF-P	Pressure/Mechanical Energy	Pressurized line opened without depressurization, rotating machinery unguarded
SIF-G	Gravity/Dropped Object/Height	Work at height without fall arrest, suspended load over workers
SIF-H	Hydrocarbon / Fire / Explosion	HC vapor cloud + ignition source, hot work without gas test, BLEVE potential
SIF-T	Toxic Gas Exposure	H2S above IDLH, SO2, benzene overexposure, NH3 release

Code	Category	O&G Examples
SIF-C	Confined Space / Asphyxiation	Oxygen-deficient atmosphere entry, engulfment risk
SIF-S	Structural / Geotechnical	Ground collapse, structural failure of platform/scaffold
SIF-V	Vehicle / Mobile Equipment**	LSEV collision, crane contact, high-consequence MVA
SIF-M	Management of Change Failure	Uncontrolled modification bypassing critical safety systems

Level 3 — Control Failure Type (Optional, for high-value reports)

Code	Control Failure
CF-ADM	Administrative control absent (No PTW, No JSA, No TBT for high-risk task)
CF-ENG	Engineering control failed or bypassed (ESD disabled, PRV bypassed, guard removed)
CF-PPE	Critical PPE absent (No BA in H2S area, no fall arrest at height)
CF-BEH	At-risk behavior (deliberate shortcut, normalization of deviance)
CF-SUP	Supervision failure (unsupervised high-risk task)
CF-MOC	Management of Change bypass

Level 4 — Confidence & Annotation Notes (Mandatory)

Field	Values	Purpose
Annotator_Confidence	High / Medium / Low	Enables filtering of uncertain annotations
SIF_Pathway_Rationale	Free text (1–2 sentences)	Critical for model explanation and SME calibration
Escalation_Flag	Yes/No	Marks report for immediate safety follow-up regardless of model use

4.2 Annotation Strategy: Reducing SME Workload by 60–70%

Full manual annotation of thousands of reports by HSE experts is not operationally feasible. The following **three-phase hybrid strategy** reduces SME hours while maintaining annotation quality:

Phase 1: Weak Supervision with Snorkel (Weeks 1–3)

Use [Snorkel](#) to programmatically generate noisy labels from expert-authored **Labeling Functions (LFs)**:

```

from snorkel.labeling import labeling_function

NEGATIVE = 0 # Non-SIF
POSITIVE = 1 # SIF Precursor
ABSTAIN = -1 # Uncertain

@labeling_function()
def lf_loto_absent(x):
    """LOTO not applied on energized equipment = SIF"""
    text = x.text.lower()
    if re.search(r'loto\s*(not|without|absent|missing|bypassed)', text):
        return POSITIVE
    if re.search(r'(not isolated|no isolation|live.*work|energized.*open)', text):
        return POSITIVE
    return ABSTAIN

@labeling_function()
def lf_h2s_above_threshold(x):
    """H2S detection above safe limits = SIF"""
    text = x.text.lower()
    if re.search(r'h2s.*(above|exceed|high|elevated|idlh)', text):
        return POSITIVE
    if re.search(r'(50\s*ppm|100\s*ppm|h2s.*alarm)', text):
        return POSITIVE
    return ABSTAIN

@labeling_function()
def lf_hot_work_no_gas_test(x):
    """Hot work without atmospheric test = SIF"""
    text = x.text.lower()
    hot_work = bool(re.search(r'(hot work|welding|grinding|cutting|spark)', text))
    no_gas_test = bool(re.search(r'(no gas test|without atmospheric|gas test not done)', text))
    if hot_work and no_gas_test:
        return POSITIVE
    return ABSTAIN

@labeling_function()
def lf_routine_housekeeping(x):
    """Housekeeping issues are Non-SIF baseline"""
    text = x.text.lower()
    if re.search(r'(housekeeping|spill cleaned|minor.*oil|trash|waste)', text):
        if not re.search(r'(h2s|toxic|explosion|fire|energized|height)', text):
            return NEGATIVE
    return ABSTAIN

# Apply all LFs and train generative model
from snorkel.labeling import PandasLFApplier
from snorkel.labeling.model import LabelModel

applier = PandasLFApplier(lfs=[lf_loto_absent, lf_h2s_above_threshold,
                                lf_hot_work_no_gas_test, lf_routine_housekeeping])

```

```
L_train = applier.apply(df=df_unlabeled)

label_model = LabelModel(cardinality=2, verbose=True)
label_model.fit(L_train=L_train, n_epochs=500, lr=0.001)
df_unlabeled['weak_label'] = label_model.predict(L=L_train)
```

Target: Develop 30–50 LFs covering all SIF categories. Aim for LF coverage > 70% of dataset.

Phase 2: Active Learning Loop (Weeks 3–8)

Train an initial classifier on weak-labeled data, then use **uncertainty sampling** to identify the most informative unlabeled examples for SME review:

```
Iteration 1: Train on 500 weak-labeled examples (seed set)
↓
Score all unlabeled reports → identify top-N most uncertain predictions
↓
SME reviews and labels top-N (uncertainty = margin < 0.15 between class
probabilities)
↓
Add confirmed labels to training set
↓
Retrain → repeat (typically 5-10 iterations until performance plateaus)
```

Uncertainty Sampling Query Strategy:

- **Least Confidence:** $1 - \max(P(y|x))$ — select least-confident predictions
- **Margin Sampling:** $P(y_1|x) - P(y_2|x)$ — smallest margin between top-2 classes
- **Entropy Sampling:** $-\sum P(y_i|x) \log P(y_i|x)$ — maximum entropy (broadest uncertainty)

Phase 3: LLM-Assisted Pre-labeling with SME Validation (Ongoing)

Use Claude or GPT-4 with a carefully engineered prompt to pre-label new incoming reports. SMEs then validate (accept/reject/correct) rather than create from scratch — reducing SME cognitive effort by ~65%.

SYSTEM PROMPT:

"You are an expert HSE analyst specializing in Oil & Gas serious injury and fatality prevention. Your task is to classify a safety observation report as either SIF Potential (1) or Non-SIF Potential (0).

A SIF Potential report describes a situation with a DIRECT CAUSAL PATHWAY to a fatality or permanent disabling injury. Consider: high-energy exposures, critical control failures (LOTO, PTW, atmospheric testing), toxic gas exposures above IDLH, and SIMOPS conflicts.

For your classification, provide:

1. Classification: SIF (1) or Non-SIF (0)
2. Confidence: High/Medium/Low

3. SIF Pathway: [1 sentence explaining the direct injury mechanism]
4. Triggering Factors: [list the specific SIF indicators found in the text]

Report: {report_text}"

Inter-Annotator Agreement Protocol:

- Minimum 2 SME annotators per report for SIF-labeled examples
- Target **Cohen's Kappa (κ) ≥ 0.75** before proceeding to model training
- Disagreement resolution: third SME arbitration or escalation to HSE lead
- All annotations logged with annotator ID, timestamp, and rationale

4.3 Feature Engineering & Preprocessing

Before tokenization, apply the following O&G-specific preprocessing steps:

Step 1: Acronym Expansion Dictionary

Build and apply a domain-specific acronym expansion dictionary *before* tokenization:

```
ONG_ACRONYMS = {
    "loto": "lockout tagout energy isolation",
    "ptw": "permit to work authorization",
    "jsa": "job safety analysis hazard assessment",
    "jha": "job hazard analysis risk assessment",
    "simops": "simultaneous operations concurrent activities",
    "bop": "blowout preventer well control",
    "h2s": "hydrogen sulfide toxic gas",
    "idlh": "immediately dangerous life health threshold",
    "esd": "emergency shutdown safety system",
    "prd": "pressure relief device safety valve",
    "scba": "self contained breathing apparatus",
    "lel": "lower explosive limit flammable gas",
    "moc": "management of change modification",
    "tbt": "toolbox talk safety briefing",
    "hw": "hot work welding cutting grinding",
    "cs": "confined space enclosed area",
    "wah": "work at height elevated work",
    "ppe": "personal protective equipment",
    "ba": "breathing apparatus respiratory protection",
}

def expand_acronyms(text: str, acronym_dict: dict) -> str:
    tokens = text.lower().split()
    expanded = [acronym_dict.get(token, token) for token in tokens]
    return ' '.join(expanded)
```

Step 2: SIF Indicator Lexicon Construction

Build a curated SIF indicator word list from SME input and incident review:

HIGH-SIGNAL POSITIVE INDICATORS (SIF):

```
["not isolated", "loto bypassed", "no ptw", "ptw expired", "without permit",
"no gas test", "atmospheric not tested", "h2s alarm", "above idlh", "LEL
exceeded",
"no fall arrest", "working at height without", "unguarded", "live circuit",
"pressurized line opened", "bop failure", "well control lost", "flame present",
"ignition source", "SIMOPS conflict", "uncontrolled release", "no JSA performed",
"confined space entry without", "esd bypassed", "relief valve blocked"]
```

HIGH-SIGNAL NEGATIVE INDICATORS (Non-SIF baseline):

```
["housekeeping", "minor spill cleaned", "signage missing", "paperwork incomplete",
"PPE worn incorrectly but corrected", "near miss no injury potential"]
```

Step 3: Negation Scope Detection

Negation handling is critical. "LOTO applied" = safety measure in place ≠ SIF precursor. "LOTO NOT applied" = SIF precursor. Apply negation scope detection:

```
# Use NegEx algorithm adapted for O&G domain
# or spaCy + negspaCy library

import negspacy
from negspacy.termsets import termset

ts = termset("en")
ts.add_upstream_negs(["without", "absent", "missing", "not", "no", "failed",
"bypassed"])
```

Step 4: Text Normalization Pipeline

```
def normalize_oag_text(text: str) -> str:
    # 1. Lowercase
    text = text.lower()
    # 2. Expand contractions
    text = contractions.fix(text)
    # 3. Normalize whitespace
    text = re.sub(r'\s+', ' ', text).strip()
    # 4. Normalize common O&G spelling variants
    text = text.replace('lock out', 'lockout').replace('tag out', 'tagout')
    text = text.replace('near-miss', 'nearmiss').replace('near miss', 'nearmiss')
    # 5. Expand acronyms
    text = expand_acronyms(text, ONG_ACRONYMS)
    # 6. Strip URLs, reference numbers (not informative)
```

```
text = re.sub(r'(https?://\S+|ref[:\s]\d+|report[:\s]#\d+)', '', text)
return text
```

5. Modeling Approach & Architecture

5.1 Baseline Models: Interpretable Foundation

A strong baseline is mandatory for two reasons: (a) it establishes a meaningful performance floor, and (b) it provides an interpretable fallback that HSE managers can audit and trust — critical for regulatory and safety-critical contexts.

Baseline 1: Keyword Rule System (Lower Bound)

Before any ML, implement a deterministic keyword-match system using the SIF indicator lexicon. This is your **floor benchmark** and serves as a quick-win deployment for stakeholder confidence:

```
def rule_based_sif_classifier(text: str, lexicon: dict) -> tuple[int, list]:
    matched_indicators = []
    text_normalized = normalize_oag_text(text)
    for indicator in lexicon['positive_sif']:
        if indicator in text_normalized:
            matched_indicators.append(indicator)
    return (1 if matched_indicators else 0, matched_indicators)
```

Expected performance: Recall ~0.65–0.75, Precision ~0.40–0.55. Insufficient for deployment, but useful baseline.

Baseline 2: TF-IDF + SVM / Logistic Regression (Primary Baseline)

```
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import LinearSVC
from sklearn.linear_model import LogisticRegression
from sklearn.calibration import CalibratedClassifierCV

# TF-IDF features: unigrams + bigrams, min_df=2 to reduce noise
tfidf = TfidfVectorizer(
    ngram_range=(1, 3),          # Trigrams capture "LOTO not applied"
    min_df=2,
    max_features=50000,
    sublinear_tf=True,          # Apply log normalization to TF
    analyzer='word',
    token_pattern=r'\b[a-zA-Z][a-zA-Z0-9_]{1,}\b' # Handles O&G alphanumeric codes
)

# SVM with probability calibration
```

```
svm_pipeline = Pipeline([
    ('tfidf', tfidf),
    ('clf', CalibratedClassifierCV(LinearSVC(class_weight='balanced',
max_iter=2000), cv=5))
])

# Logistic Regression (more interpretable, coefficient analysis available)
lr_pipeline = Pipeline([
    ('tfidf', tfidf),
    ('clf', LogisticRegression(class_weight='balanced', C=1.0, max_iter=1000))
])
```

Baseline Expected Performance (estimated): F1 ~0.68–0.75, Recall ~0.78–0.84

Interpretability Bonus: Extract top-weighted TF-IDF features per class — this list *becomes* your initial SIF indicator lexicon, validated by data rather than assumptions.

5.2 Advanced Transformer-Based Architecture

Stage 1: Domain Language Model Selection

Do NOT start from generic `bert-base-uncased`. Start from the closest domain-available pre-trained model and fine-tune progressively:

Model	Pre-training Corpus	Recommended For	Notes
<code>bert-base-uncased</code>	BookCorpus + Wikipedia	Not recommended as starting point	Too generic
<code>roberta-base</code>	Diverse English text	Better baseline than BERT for classification	More robust training
<code>SciBERT</code> (allenai)	Scientific papers	Good for HSE if no better option	Vocabulary overlap with HSE standards
<code>SafetyBERT</code> (custom)	O&G safety corpora	Best starting point if buildable	Pre-train on OSHA+BSEE+HSE+PHMSA data
<code>xlm-roberta-base</code>	100-language corpus	Required for OIL (Hindi/Assamese reports)	Multilingual coverage

Stage 2: Three-Phase Fine-tuning Strategy

Phase 1: Domain Adaptive Pre-Training (DAPT)

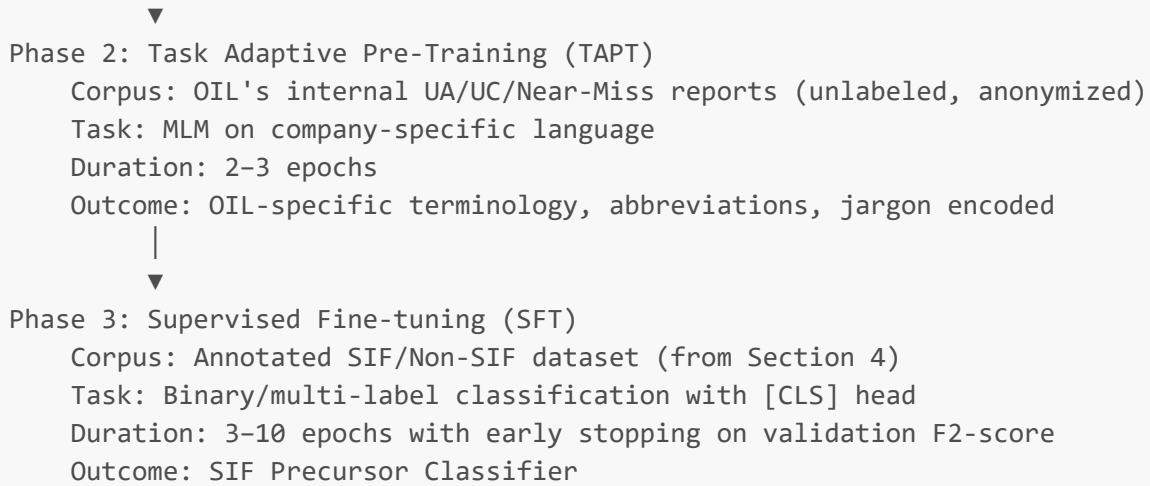
Corpus: OSHA SIR + BSEE + PHMSA + UK HSE + CSB reports (all anonymized)

Task: Masked Language Modeling (MLM) on O&G safety text

Duration: 3–5 epochs on full corpus

Outcome: O&G safety language representation

|



Stage 3: Model Architecture Specification

```
from transformers import (AutoModelForSequenceClassification,
                          AutoTokenizer, TrainingArguments, Trainer)

model_name = "xlm-roberta-base" # For OIL (multilingual)

# Classification head configuration
model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2, # Binary: SIF / Non-SIF
    problem_type="single_label_classification",
    hidden_dropout_prob=0.2, # Regularization – important for small
    datasets
    attention_probs_dropout_prob=0.2
)

training_args = TrainingArguments(
    output_dir="./sif_classifier",
    num_train_epochs=8,
    per_device_train_batch_size=16,
    warmup_steps=200,
    weight_decay=0.01,
    learning_rate=2e-5, # Standard for BERT fine-tuning
    evaluation_strategy="epoch",
    save_strategy="best",
    metric_for_best_model="recall", # Optimize for recall, NOT accuracy
    load_best_model_at_end=True,
    fp16=True, # Mixed precision for speed
)
```

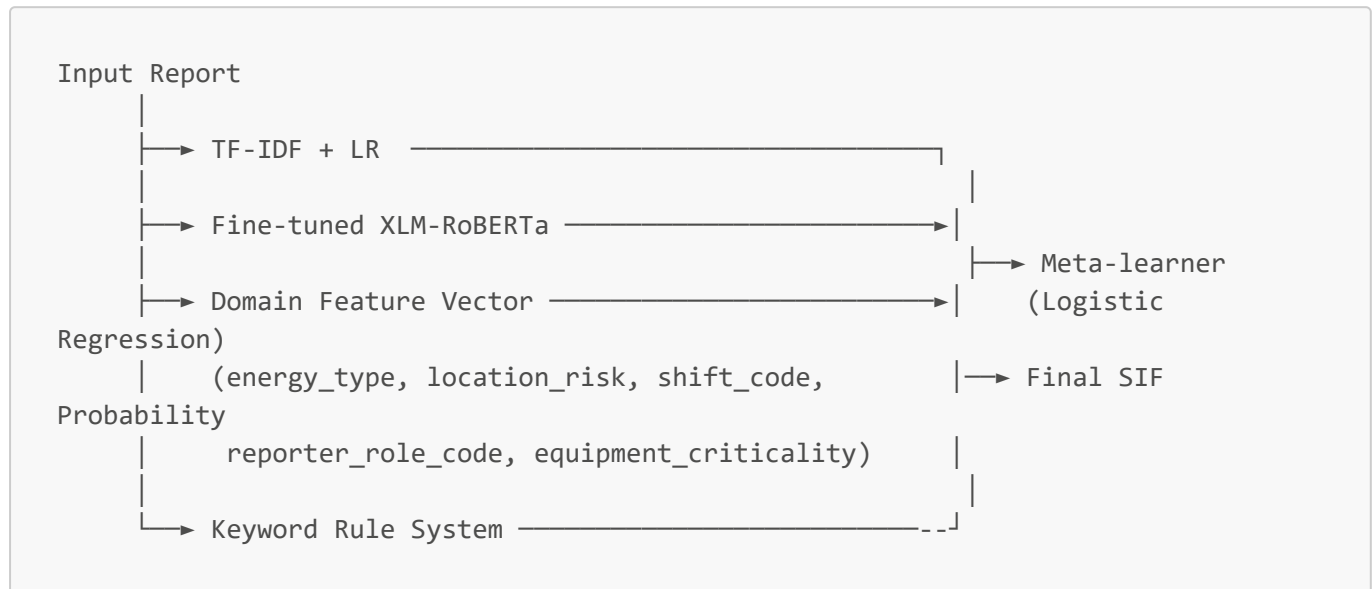
Long-Document Strategy

Many O&G incident investigation reports exceed the 512-token BERT limit:

- **Short reports (< 512 tokens):** Standard BERT/RobERTa — no modification needed

- **Medium reports (512–1024 tokens):** Sliding window with stride; aggregate predictions
- **Long reports (> 1024 tokens):** Use **Longformer** (4096 token context) or hierarchical model (sentence encoder → document encoder)
- **Practical recommendation:** For most UA/UC cards (typically 50–250 words), standard 512-token limit is sufficient. Investigate reports may need Longformer.

Ensemble Architecture (Production Recommendation)



The **domain feature vector** encodes non-text structured features:

- Location risk code (based on asset hazard classification)
- Time/shift (night shifts have higher SIF potential statistically)
- Reporter role (operator vs. supervisor vs. contractor)
- Equipment criticality tier (from asset register)

5.3 Train/Validation/Test Splits — Data Leakage Prevention

Critical: Random splitting WILL cause data leakage in time-series safety data. A model trained on 2022 data and tested on 2021 data is not predicting SIF precursors — it is memorizing historical patterns in reverse.

Mandatory: Temporal Splitting

Training Set:	Reports from Year 1 through Year (N-2)	~70% of data
Validation Set:	Reports from Year (N-1)	~15% of data
Test Set:	Reports from Year N (most recent year)	~15% of data

Example for OIL:

Training:	2015–2022	(all historical)
Validation:	2023	(recent, for hyperparameter tuning)
Test:	2024	(held-out, final evaluation – NEVER touched during development)

Additional Leakage Prevention Rules

- **No report duplication across splits:** Some EHS systems log corrections as new entries — deduplicate by canonical incident ID before splitting
- **Location stratification:** Ensure each field location (Duliajan, Jorhat, Rajasthan fields) is represented in all splits
- **SIF class stratification:** Given rarity, ensure test set contains statistically sufficient SIF positives (minimum 50 confirmed SIF examples in test set)
- **Cross-validation within training set:** Use time-series cross-validation (expanding window) — NOT k-fold

```
from sklearn.model_selection import TimeSeriesSplit

tscv = TimeSeriesSplit(n_splits=5, gap=30) # 30-day gap between train/val to prevent leakage
for train_idx, val_idx in tscv.split(X_train_temporal):
    # Train on expanding window, validate on next window
```

6. Evaluation Metrics & Business Integration

6.1 Why Recall Dominates: The Asymmetric Cost Function

In safety-critical classification, the standard accuracy metric is not only inadequate — it is **actively misleading**. The evaluation framework must be explicitly designed around the **asymmetric cost of classification errors**:

Error Type	Classification Term	Real-World Consequence	Estimated Cost (O&G context)
False Negative (FN)	Miss a SIF precursor — model labels it Non-SIF	Undetected precursor → potential SIF event	Fatality: \$7M–\$30M+ (OSHA VSL) + regulatory sanctions + reputational damage + human tragedy
False Positive (FP)	Alert on a Non-SIF report	Safety officer investigates unnecessary report	~2–4 hours of safety officer time (\$200–\$400)

Cost Ratio: The cost of one False Negative ≈ 10,000–100,000× the cost of one False Positive.

Consequence: Optimize aggressively for **Recall (Sensitivity)**. Accept lower Precision. Accept higher alert volume. **Never miss a real SIF precursor.**

Primary Evaluation Metrics

Metric	Formula	Target	Rationale
Recall (Sensitivity)	$TP / (TP + FN)$	≥ 0.92	Primary business metric — must not miss SIF events

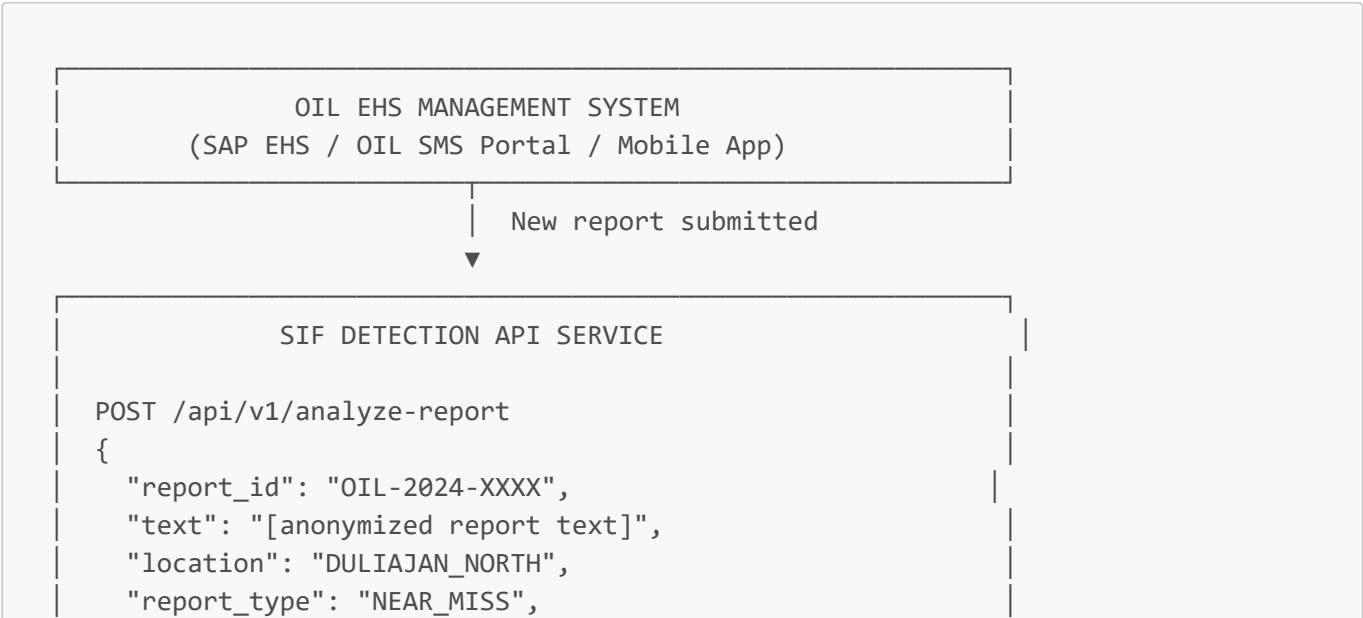
Metric	Formula	Target	Rationale
F2-Score	$5 \cdot P \cdot R / (4 \cdot P + R)$	≥ 0.75	Weights Recall 2× over Precision — operational compromise
PR-AUC	Area under Precision-Recall curve	≥ 0.70	More informative than ROC-AUC for imbalanced datasets
Specificity	$TN / (TN + FP)$	≥ 0.80	Prevent alert fatigue — too many FPs paralyzes safety staff
ROC-AUC	—	≥ 0.88	Overall discriminative ability
ECE (Expected Calibration Error)	Calibration measure	< 0.05	Probability scores must be trustworthy for risk ranking

Do NOT use Accuracy as a primary metric. A model predicting all Non-SIF achieves 95% accuracy on a 5% SIF-prevalence dataset. This is the baseline failure mode.

Secondary Business Metrics

Metric	Description
Alert Conversion Rate	% of model-flagged reports that SME investigation confirms as genuine SIF precursors — tracks Precision in operational context
Mean Time to Alert (MTTA)	Time from report submission to safety officer notification — target < 5 minutes
SIF Incident Rate (Post-deployment)	Lagging indicator — did confirmed SIF events decrease after system deployment?
Near-Miss Reporting Rate	Leading indicator — did field reporting increase (suggesting increased safety culture engagement)?

6.2 Deployment Architecture for Real-Time Alerting



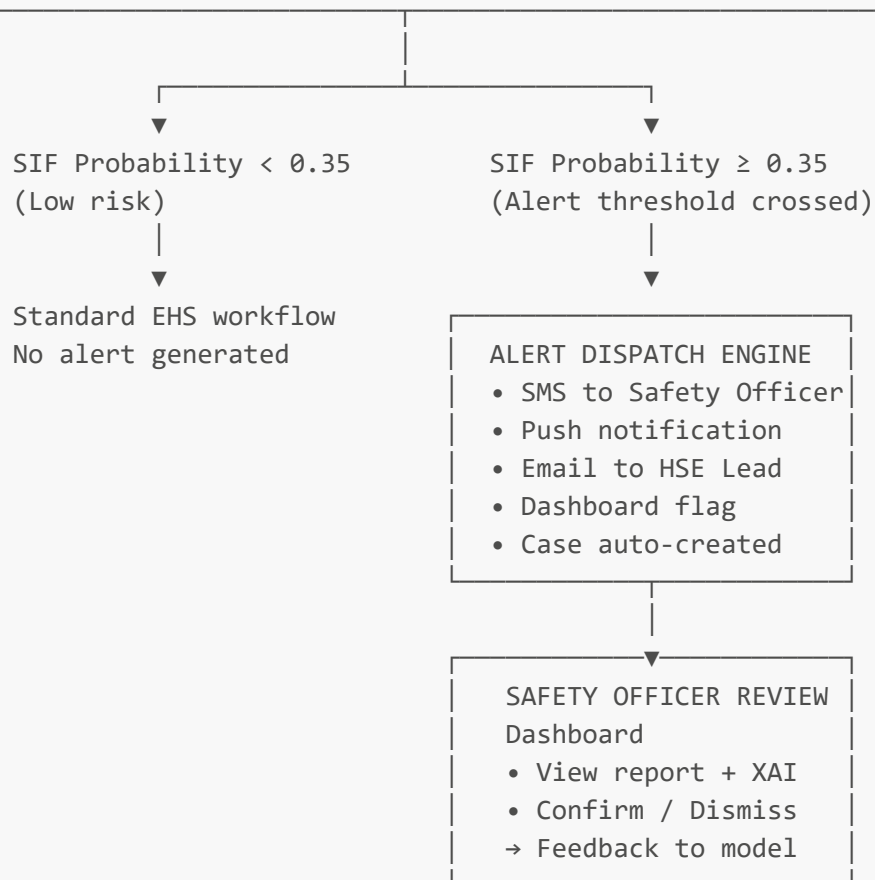
```
"equipment_id": "[asset_tag]"
}
```

Processing Pipeline:

1. Text normalization + acronym expansion (< 50ms)
2. Ensemble model inference (< 500ms)
3. Threshold evaluation + confidence scoring
4. XAI explanation generation (LIME/SHAP)
5. Response + alert trigger decision

Response:

```
{
  "sif_probability": 0.87,
  "classification": "SIF_POTENTIAL",
  "confidence": "HIGH",
  "sif_categories": ["SIF-H", "SIF-ADM"],
  "triggering_factors": [
    "hot work detected without PTW",
    "atmospheric test absent",
    "hydrocarbon equipment proximity"
  ],
  "recommended_action": "IMMEDIATE_STOP_WORK",
  "alert_recipient": "SAFETY_OFFICER_NORTH_FIELD"
}
```



Continuous Learning Loop

The deployment must include a **model monitoring and retraining pipeline**:

Monthly Cadence:

1. Collect safety officer feedback (confirm/dismiss decisions)

2. Extract new confirmed SIF reports → add to training set

3. Monitor: concept drift via PSI (Population Stability Index) on feature distributions

4. Retrain trigger: if Recall on validation set drops below 0.90 threshold

5. Champion/Challenger deployment: shadow new model on production traffic for 2 weeks

6. A/B evaluation: promote challenger if F2-score improvement ≥ 2%

Explainability (XAI) — Non-Negotiable for HSE Deployment

Safety officers will **not trust a black-box model** that flags reports with no explanation. Implement:

```
# SHAP for transformer models
import shap
explainer = shap.Explainer(model, tokenizer)
shap_values = explainer([report_text])

# LIME for token-level attribution
from lime.lime_text import LimeTextExplainer
explainer = LimeTextExplainer(class_names=['Non-SIF', 'SIF'])
exp = explainer.explain_instance(report_text, model.predict_proba,
num_features=10)

# Output: "Model flagged this report because these phrases strongly indicate SIF:
#         'without ptw' (+0.43), 'hot work' (+0.31), 'hydrocarbon line' (+0.28)"
```

Appendix A: Recommended Technology Stack

Layer	Technology	Purpose
Data Lake	AWS S3 / Azure Blob + Delta Lake	Raw + processed data storage
Data Versioning	DVC (Data Version Control)	Reproducible dataset versions
ETL Orchestration	Apache Airflow / Prefect	Pipeline scheduling
NLP Framework	HuggingFace Transformers + spaCy	Model training & inference
Weak Supervision	Snorkel (snorkel.ai)	LF-based labeling
Annotation Platform	Label Studio / Prodigy	SME annotation interface
Experiment Tracking	MLflow / Weights & Biases	Model versioning, metric tracking
Model Serving	FastAPI + Docker + Kubernetes	Production API
Monitoring	Evidently AI / Arize	Data drift, model performance

Layer	Technology	Purpose
XAI	SHAP + LIME	Prediction explanations
PII Anonymization	Microsoft Presidio + spaCy	Data sanitization
OCR	AWS Textract / Tesseract 5.0	Handwritten/scanned report digitization

Appendix B: Project Timeline (Indicative)

Phase	Duration	Key Deliverable
Phase 0: Data Discovery	2–3 weeks	Data inventory, access agreements, PII framework approval
Phase 1: ETL & Sanitization	4–6 weeks	Cleaned, anonymized dataset in data lake
Phase 2: Annotation Sprint	6–8 weeks	2,000+ annotated reports ($\kappa \geq 0.75$)
Phase 3: Baseline Modeling	2–3 weeks	TF-IDF+SVM baseline, keyword system
Phase 4: Transformer Fine-tuning	4–6 weeks	Fine-tuned XLM-RoBERTa SIF classifier
Phase 5: Evaluation & XAI	2–3 weeks	Full evaluation report, explainability integration
Phase 6: Pilot Deployment	4–6 weeks	Shadow deployment at one OIL field location
Phase 7: Full Rollout	8–12 weeks	Production deployment + monitoring pipeline
Total	~6–9 months	—

Research Synthesis Version 1.0 | Prepared for OIL SIF Precursor AI/NLP Engine Project
Classification: Internal Research Use | Review before external sharing